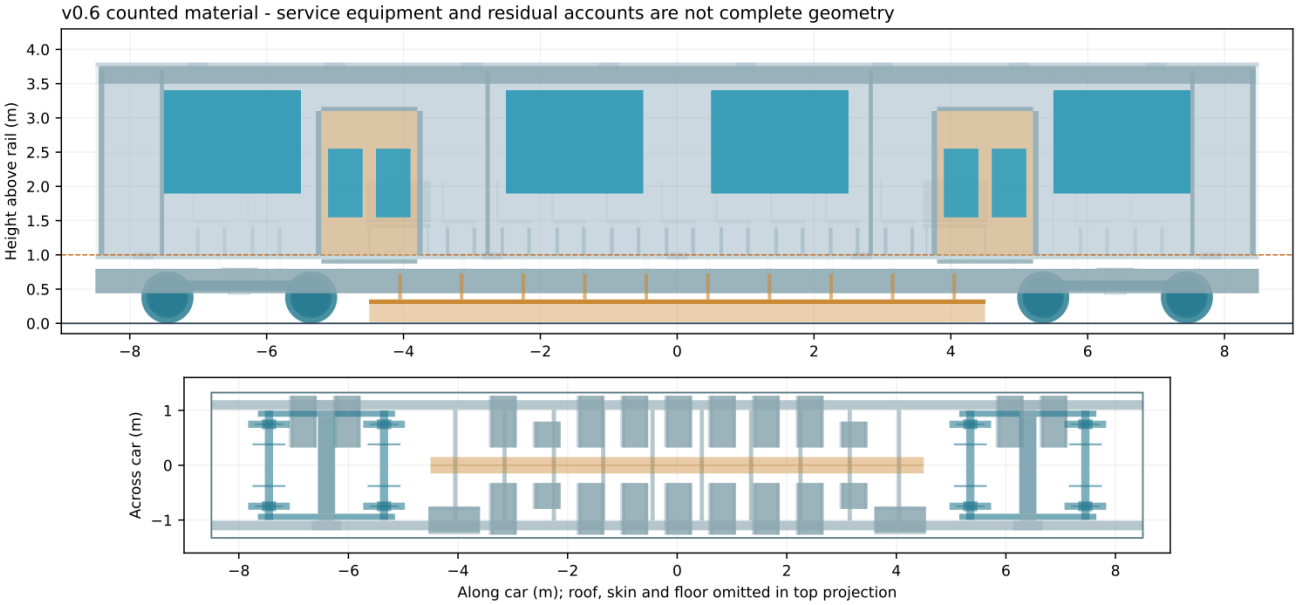
OpenSourceRail

v0.6 - Station-powered freewheeling rail

Simple vehicles. Repeated civil works. Comfortable, frequent journeys.

Reference engineering and prototype design | 26 October 2026



Counted material proposal. Supplier reference assemblies and residual accounts are not complete installed geometry.

Six lightweight cars run on passive wheel bogies and conventional steel rails. Wayside linear-induction motors launch and receive the train through passive onboard aluminium secondaries. The train has no traction motors, traction battery or suspension magnets.

Selected for development, with explicit trade-offs. Physical operation, lowest whole-life cost, complete installed mass and manufacturing release remain unestablished.

1. Preserve the ethos; change the support

OpenSourceRail aims to reduce whole-life transport cost through repeated components, local manufacture, accessible repair and open engineering. v0.6 removes magnetic levitation and its continuous steel-facing assemblies. This does not remove the need to prove station-only propulsion, safe stopping or the cost of the complete railway.

Location	Reference architecture
Train	Six 17 m aluminium/GFRP comfort cars; identical passive bogies, independent brakes, ATP, communications, service battery, HVAC and solar roof.
Ordinary guideway	Conventional running rails and direct fastenings on precast concrete; box, twin-beam and U-beam alternatives.
Stations	Finite LIM primary sections, inverters, power/charge equipment, reserved receiving berth and compact elevated platforms.
Control	Existing station-led GoA4 architecture with continuous onboard ATP and independent mechanical braking. No mainline switches; depot exceptions only.

Each car retains 44 fixed seats plus two wheelchair places, zero planned standing and two doors on one side. Four-abreast seating and door-zone wall benches preserve the comfort choice. 276 people/train at 120 s gives 8,280 passengers/hour/direction; this is capacity arithmetic, not a demonstrated protected headway.

The revised floor reference is 1.0 m above rail to clear conventional running gear. Platform access, one-sided doors, centre-platform compatibility, evacuation and passenger admission need physical verification. The 160-person/car crowd case remains a structural and braking sensitivity.

2. Equipped vehicle and mass accounts

Per car	Reference result	Meaning
Accounted tare	19.222 t	Materials, supplier references and unlocated accounts
46-person normal load	22.672 t	75 kg/person hypothesis
160-person crowd control	31.222 t	Not the intended normal capacity
Located material	9.132 t	Native geometry subset, not complete tare
Installed tare / CG / inertia	Unknown	Residual stock and complete equipment placement unresolved

Twelve two-axle bogies give 24 axles and 48 wheels. Motors and gearboxes are absent. Suspension, bearings, disc brakes, WSP, parking and wheel odometry remain. The 6,730 kg/car bogie reference includes partial proposed material and a transparent supplier/detail residual.

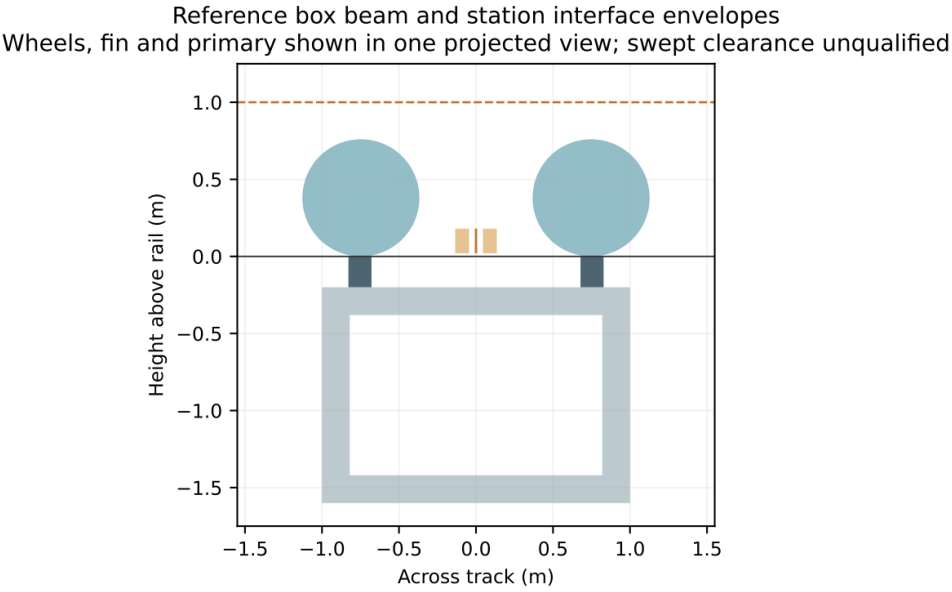
The new body has concentrated bogie reactions at 12.8 m centres rather than distributed magnetic supports. Deep side sills and the retained roof longitudinal members provide the intended body load path. The isolated side-sill fixture fails its declared stiffness limit. Ideal whole-body chord coupling is a necessary upper bound; posts, joints, shear transfer, openings, torsion, fatigue and crash behaviour remain unqualified.

Inherited body/service residuals may overlap new battery and electrical equipment. No unsupported mass subtraction is made. This conservative account needs disaggregation before it becomes a definitive equipped mass. Split battery cassettes, connections and removal paths remain unresolved.

The revised roof proposes HP60 core and a white finish. Glass, doors, floor and seating retain source ancestry; plastics remain alternatives requiring impact, fire, optics, retention and repair evidence.

3. Station motor: requirements before claims

The reference is a double-sided LIM. Powered primaries and converters remain wayside; each train car carries a 9 m passive aluminium secondary with support and heat-spreading stock. Single-sided LIM and synchronous PM alternatives remain comparisons.



Reference projected interface. Nominal 15 mm side clearance is a design target, not a demonstrated dynamic motor gap.

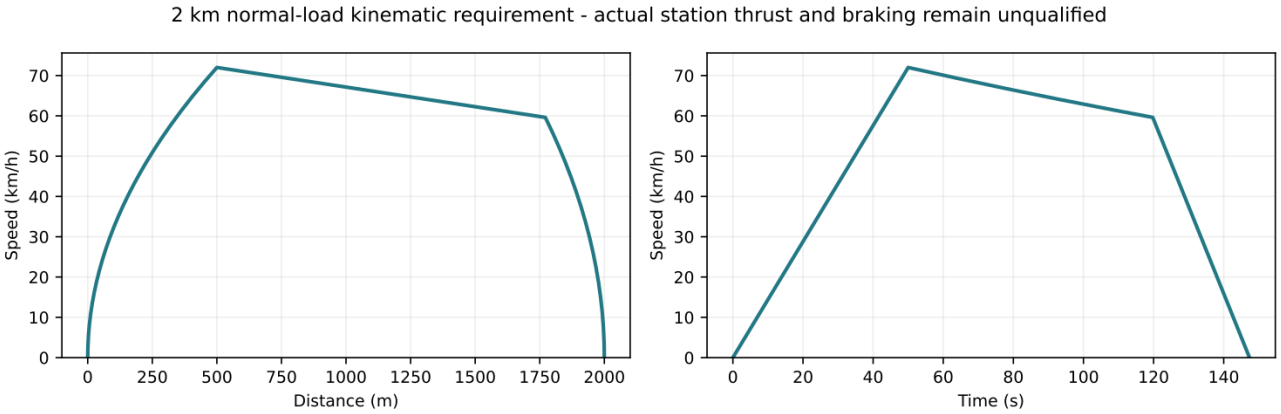
2 km normal-load requirement	Calculated target
Launch acceleration / speed cap	0.4 m/s2 /20 m/s; ideal powered distance 500 m
Shared winding library choice	900 m across launch and receipt
Section addresses / independent branches	72 per direction; at least 2 train current/phase branches
Actual available thrust / voltage / current	Unknown; requires actual primary and moving-secondary field maps

The executed GetDP manufactured conductive-slab model converges against an exact imposed travelling-field solution. It verifies a numerical method. It omits the actual primary, steel, slots, finite-height current closure, motor ends and inverter circuit; conditional force is not available train thrust.

Moving suspension, wheel wear, alignment, fin expansion and unwanted motor normal forces must fit the same clearance envelope. Full entry, exit, sectional handoff and 120-second repeated duty remain gates.

4. Reference motion and independent stopping

Input/result	Reference
Route / load	2,000 m, six normal-load cars; flat route and declared resistance
Runtime	147.362 s analytical target
Berth resource interval	85.46 s necessary screen; 120 s target unqualified
Resistance	0.0025 mg +50 N/car +1.8 v2 N/car hypothesis; not measured drag
Motion evidence	Required force, energy and geometry; physical train journey not completed



Target acceleration, natural coasting and receiving brake profile. Prescribed kinematic acceleration does not prove motor capability.

Empty, normal and crowd-control 1.5/2 km cases are evaluated, with grade and headwind sensitivities. The longest retained adjacent city gap is approximately 3 km. Adverse uphill cases can stop before receipt; downhill cases can violate the unpowered speed cap. Failures stay visible instead of adding silent boosters.

Onboard mechanical service/emergency/parking brakes must work independently of station power. A 0.05 adhesion hypothesis on a 2% downhill gives about 700 m stopping from 20 m/s including 1 s response, with no regeneration or resistance credit. Actual wet/hot curves, WSP, brake channels and thermal duty remain unknown.

5. Service energy, solar and hot weather

The auxiliary pack is for HVAC, doors, lights, communications, ATP and emergency functions only. No wheel-drive connection exists. The 82.8 kWh reference uses 75 industrial Type3-20 modules, 25 series by 3 parallel, plus enclosure, cooling, conversion, protection and retention accounts: 1,345 kg/car. This is not a railway-qualified installed pack.

Necessary energy/current check	Result and qualification boundary
35 s station charge target	200 kW/car; 1.2 MW/train before simultaneous traction and station services
Normal hotel load	35 kW/car declared reference; prototype repeated-energy bound passes
Adverse hotel load	50 kW/car requires extra charge time in the 2 km case
Hot-weather battery	Catalogue module ambient upper limit 45 C; 50 C surroundings require a proven cooled enclosure
Ageing/reserve	80% end-of-life and 90% hot-capacity hypotheses; SOC window 20%-90%
Solar	Flush roof PV retained; no unproven energy yield credited to make service pass

Catalogue voltage and continuous-current bounds do not establish charge acceptance versus SOC, temperature or life. A missed charge, stranded train and independently protected essential reserve require dispatch and emergency-energy validation. Longer/hot legs can change terminal charging, fleet and service.

LIM secondary heating is exposed over several slip assumptions. Active-plate-only and full-aluminium heat-capacity bounds differ; actual thermal conductance, repeated-duty cooling and receiving losses are unknown. A large calculated force obtained with imposed field can carry unacceptable secondary heat.

HVAC delivered cooling and COP at 50 C remain unknown. Passenger ventilation, solar gain, stationary heat rejection, redundancy, roof hot strength and durable reflective coating must close together.

6. Three civil alternatives and construction

Conventional rails replace continuous magnetic facings. Compare a closed box, two narrow beams with gauge-control diaphragms, and a U-section. Lower concrete area alone does not establish the least-cost guideway: torsion, rail support, foundations, containment, maintenance and erection matter.

Selected 20 m crowd-sized box	Counted quantity
Concrete	22.282 m3
Running rails	2.408 t
Steel incl. preliminary rebar/tendons	6.448 t
Actual installed cost	Unknown

The selected reference is 20 m with 26 prestressing strands sized to the crowd load. The same fixed beam/tendons are checked at empty, normal and crowd loads; three families across four spans retain their separate comparisons. It is a necessary longitudinal screen, not an installed-cost winner.

Moving axle-load envelopes are calculated at each longitudinal section. Each section receives its own critical train position; it does not reuse one global peak. Prestress, no-train and release stress checks and a conservative deflection envelope remain analytical screens.

Steel boundaries differ from v0.5: its 21.627 t/25 m interface account excluded reinforcement/prestress, while the new total includes preliminary detailing. Neither includes complete piers, foundations, bearings, walkways, containment and station primaries. Concrete quantity can increase. No steel saving is converted directly to a dollar saving.

Construction sequence

Survey and geotechnical investigation; foundations; piers/headstocks; bearings and temporary stability; precast lift; rails and gauge setting; station motor installation; survey, inspection and independent release. Actual ground, crane radius, labour, factory throughput and supplier lead times are required.

Same-candidate BOM, manufacturing, procurement, assembly and controls records are research drafts. Unknown prices, availability dates and material release are not replaced with zeros or unconstrained schedules.

7. Automation, faults and retained geography

Station-led GoA4 reserves route and receiving berth, proves configuration and train readiness, sequences launch equipment and supervises receipt. Independent onboard ATP continuously enforces authority. Stations never replace the train brake protection layer.

Actual Rust ATP/interlocking, TCMS, doors and SCADA regressions are retained. New configuration-bound wheel/station predicates exercise readiness faults and mechanical brake requests. Inputs are fixtures: they establish neither a physical brake curve nor a safety-qualified GoA4 product. Historical magnetic healthy-gap fixtures remain archived.

Fault	Required independent response
Receiving drive or communications lost	Respect authority and independent stopping envelope; cut station thrust and command brakes where required
Localisation/integrity uncertain or berth occupied	Restrict authority and stop; no blind timer-based launch
Wet rail or brake channel loss	Qualified WSP and degraded brake floor; protect the larger stopping distance
Off-zone stop	No autonomous restart; protected rail-compatible rescue/towing and evacuation plan
Service power lost	Independent essential/brake supply and emergency ventilation; hot-weather endurance qualification

The 266 city geographies and detailed maps are preserved. No v0.6 city performance packages are generated. A metadata-only scan covers 17,268 adjacent operating-stop gaps; the longest is 2,999.931 m in Faisalabad. Grades, sites, demand, transfers, power and prices remain city-specific.

City commands return an explicit evidence hold. Existing fleet, winding, cost and finance figures describe historical designs. They are not carried over as v0.6 results.

8. Equal service and the selection decision

Candidate	Accounted/proxy tare	Normal comparison load
v0.6 wheel/LIM	19.222 t/car	22.672 t/car
v0.5 PM maglev	17.357 t/car	20.807 t/car
v0.4 station-launch mass proxy	23.680 t/car	27.130 t/car

All comparison profiles use 276 people/train, 75 kg/person, 120 s headway, 72 km/h cap and the same nominal acceleration/braking. Drag hypotheses remain architecture-specific and unmeasured. The v0.4 row is an optimistic arithmetic proxy from its 16.5 m LM3, with traction/battery categories removed before replacement service equipment; it is not a designed 17 m comfort vehicle.

v0.6 is selected for research and prototype development by user direction because wheel support removes the unresolved levitation system and continuous magnetic-facing assemblies. Its vehicle account is heavier than v0.5; station-only LIM, high platforms and rescue introduce costs. The selection deliberately records that trade-off.

Whole-life economic acceptance remains unknown. Vehicle and station prices, supply reinforcement, construction, maintenance, wheel/rail renewal, energy tariffs, reliability, fleet, staff and financing need equal-service source data. The current package does not demonstrate that v0.6 beats v0.5 or a simpler powered rail alternative.

Release remains separate

A prototype package needs reviewed drawings, tolerances, joining, tooling, BOM, procurement, inspection and an instrumented test programme. Development adoption does not grant manufacturing, physical motion or operating release.

9. Evidence gaps and next work

Gate	State	Required evidence
installed-vehicle	unknown	Locate and weigh all residual stock, joinery, bogie and service assemblies; obtain complete CG and inertia.
isolated-underframe-stiffness	fail	The isolated hollow side-sill fixtures exceed the declared deflection limit. Whole-body load sharing through retained longitudinal members, posts and joints must be demonstrated or redesigned.
body-and-bogie-structure	unknown	Concentrated bogie support changes the body load path. Local/frame fixtures do not establish joined-body stress, buckling, fatigue, crash, modes or bogie qualification.
wheel-rail-dynamics	unknown	Qualify wheel profiles, hunting, curves, suspension, ride, flange forces and derailment containment.
station-LIM-force-map	unknown	No accepted thrust/current/frequency/gap/temperature map for the complete station drive and moving secondary; targets are required forces.
station-electrical-thermal	unknown	Qualify inverters, section handoff, supply, slip/secondary heat, unwanted normal forces and 120-second repeated duty.
independent-braking	unknown	Validate supplier brake/WSP/adhesion/thermal curves, channel failures, parking, receiving-power loss and degraded gradients.
hot-weather-services	unknown	HVAC delivered capacity and COP at 50 C, roof long-term structural temperature and actively cooled service pack remain unqualified.
coupled-journey	unknown	Reference motion is analytical. Required physical drive/brake maps, wheel dynamics and flexible-body model are not available for a qualified six-car trip.
GoA4-protection	unknown	Existing Rust ATP/interlocking and new wheel interface predicates require candidate-correlated braking, sensing, timing, communications and fault evidence.
civil-and-site	unknown	Alternative sections are necessary screens. Prestress, fatigue, torsion, lateral/derailment loads, foundations, survey, erection and independent review remain open.
rescue-and-evacuation	unknown	Motorless off-zone restart needs a protected rescue/tow system, emergency access and an acceptable high-viaduct evacuation plan.
whole-system-economics	unknown	No same-service supplier quotes or complete CAPEX/OPEX/renewal comparison; lower installed cost has not been established.
city-regeneration	hold	Preserve 266 geographic inputs. Numerical city publication is held pending reference propulsion, braking, route and economic evidence.

10. Native motor, body and brake follow-up

The bounded follow-up adds 18 current-excited GetDP motor cases and 24 actual Chrono brake executions. These are component/reduced-model experiments, not a physically coupled launch-coast-arrival journey.

Executed comparison	Result
Full 12.5 m primary and 9 m secondary	7.174 kN/car versus 11.103 kN required
Peak steel field	2.240 T exceeds declared 1.3 T linear-domain bound
Bare primary copper and steel	228.2 kg/m, excluding cooling, supports and power equipment
Full motor heat at tested point	52.1 kW secondary plus 43.6 kW primary copper per engaged module

Successful zero-current, phase-reversal and energy checks do not repair the force deficit or out-of-domain steel. Winding regrouping can reduce an arithmetic voltage requirement at much higher converter current, but finite-overlap current sharing and thermal duty are unknown. Installed available force stays unknown.

Normal-load braking from 20 m/s	5 ms Chrono result
dry	329.6 m
wet-downhill	719.0 m
one-channel-lost	1399.4 m
insufficient-adhesion	Cannot stop

Wet/downhill uses adhesion 0.05 on a 2% downgrade; lost channel retains four discs/car. Assumed pad friction, piston pressure and ideal instantaneous WSP are not qualified hardware. The crowd failed-channel case needs over 3 km. Fault detection, authority, route speed and protected overrun must use a validated degraded stopping envelope.

Independent body verification rejected native rotational beam equations and an overly rigid solid-section fixture. Rejected results remain recorded. Counted post connections and actual floor cutouts/collars add 77.154 kg/car before unknown joining hardware; they are an unadopted alternative. Native stock fit-up does not prove body strength, fatigue or complete assembly.

Counted wheel, axle and disc rotation adds 1,770.387 kg/car equivalent inertial mass, with no added static weight. It now feeds launch work, coasting and brake energy. Exact findings, benchmark dispositions and compact source-bound evidence are in docs/design-engine/v06-physical-closure/.

11. Failure-driven candidate generations

The existing runner completed 3 generations and 6 physical wheel/LIM candidates. Code generated each revision from recorded failures; the reference geometry was not manually edited between trials.

Generation /candidate	Accounted tare/car	Normal deflection	Motor validity
g0 / 190290ff88	19.299 t	111.1 mm	invalid
g0 / b96ae12040	19.306 t	32.2 mm	invalid
g1 / 035b98536f	20.231 t	27.8 mm	invalid
g1 / 7147932737	19.324 t	32.3 mm	invalid
g2 / 168c9dc7fe	20.764 t	22.7 mm	invalid
g2 / d8941f800e	19.324 t	32.3 mm	invalid

The next experiment accounts for 20.764 t/car. Its crowd deflection is 33.6 mm against 25.6 mm. Required thrust is 11.758 kN/car; the conditional calculation is 9.733 kN. Invalid motor fields cannot establish available thrust or physical feasibility.

The lighter participating-panel branch remains laterally nonconverged. Its vertical values are diagnostic. Synthetic laminate/core constants and ideal clamp transfer are unqualified; invalid body results trigger model work rather than unsupported geometry revisions.

Valid frame failures generate deeper/thicker counted stock. Motor revisions change teeth, backiron and windings; matched fin/neck geometry, car mass, rotor energy, brakes and civil loads propagate together. The fixed linear-steel limit cannot be relaxed to manufacture a pass.

Exact dependency digests, source revisions and solver identities bind caches and restart. The Pareto comparison ranks diagnostic priority and counted quantities. Prices and complete installed equipment remain unknown; research selection cannot satisfy integrated acceptance.

Every candidate generates native FreeCAD/STEP, reopened IFC, material takeoff and draft assembly, QA, procurement and maintenance records for known stock. Unknown equipment has no placeholder solids. Position-dependent journeys, joint/fatigue checks, qualified brakes and complete mass/CG/inertia remain open.

Inspect docs/design-engine/v06-optimisation/ and ./osr design campaign create/run/compare/publish. Typed next tasks retain candidate and trigger IDs with bounded scope. The 266 city packages and prior physical evidence remain preserved.

12. Challenge the railway architecture

The umbrella research campaign executes 509 analytical cases across 174 physical definitions and 2 generations. It preserves the selected v0.6 reference while testing different propulsion distribution, powered-track length, speed, formation and concrete sections.

Family	Research arrangement
P1	Current station LIM
P2	Lower-speed station LIM
P3	Sparse wayside boosters
P4	Creep/recovery drive
P5	Distributed wayside propulsion
P6	Onboard wheel traction
P7	Alternative passive secondary

Required-force trajectories compare 4/6/8 cars, 1.5/2/3 km station spacing and 43.2/57.6/72 km/h caps on a 6 km comparison ring. New drive and recovery equipment remains unweighed. Changed station density changes access service. Protected headway, actual available thrust and complete installed costs remain unknown; no economic winner or physical journey is claimed.

Six-axis finite-area joint fixtures now check work, equilibrium, reciprocity and modal refinement. The counted whole-body alternative adds 113.008 kg/car of known stock. After a retained 3 GiB allocation failure, a separately recorded 5 GiB resource revision completes the finer mesh without changing physics or limits. The vertical mesh change is 1.43%; normal deflection 28.018 mm exceeds the provisional 25.6 mm comparison. The new adaptive policy treats that comparison as advisory. Perfect bonds and ideal section transfer require physical calibration and local joint/strength/fatigue checks.

The nonlinear DC magnetic method reproduces a generic 49-point material circuit and its independent coenergy-gradient force. It is not selected motor steel or a moving nonlinear LIM. Spatial entry/centre/exit controls remain linear. Full nonlinear moving thrust, losses, inverter and thermal maps are still needed before a motor can supply a physical journey.

Capability admission binds geometry, interfaces, evaluator, benchmark, source and validity domain. Unsupported configurations generate bounded implementation or measurement tasks. M1 component correlation, M2 passive-carriage launch/coast/stop and M3 integrated pilot plans require measured data, uncertainty and independent review; none is marked complete.

Inspect docs/design-engine/v06-architecture/. Run ./osr design architecture run/publish/verify. Exact-source caching and an interrupted-run receipt demonstrate restart. The Workbench exposes all seven families, and scheduled CI runs bounded joint and nonlinear magnetic method checks.

13. Adaptive design from broad requirements

The function-first generator declares 102 extensible variables across 14 domains. Materials, honeycombs, layups, load paths, joints, support arrangements, propulsion, fitout, service and civil/construction methods can change under separately identified research configurations. Examples are seeds, not an exhaustive menu.

Mandatory functional and safety obligations remain. Quantitative hard limits need a justified interface/material/function basis. Initial dimensions, L/500-style deflection comparisons, speed and headway are explicitly revisable hypotheses or targets. Numerical-model limits stay separate: changing the old linear-steel field bound cannot create nonlinear physics.

Native section experiment	Tare account/car	Normal bending	Stock change
180 x600 x5 mm	19.287 t	19.793 mm	-48.28 kg
180 x500 x6 mm	19.316 t	22.334 mm	-19.17 kg

Changes are relative to the previous finite-area jointed body: 19.335 t/car and 28.018 mm normal displacement. These are counted stock plus inherited accounts, not a complete installed vehicle mass. Normal coarse/fine and crowd runs use the actual new section stock and passenger loads. Both displacement trends are checked; each crowd result has only one mesh.

The 600 mm-deep, 5 mm-wall experiment redistributes stock: about 48 kg/car less material and 19.793 mm normal bending. A 500 mm-deep alternative retains more underbody space. These model observations motivate further design work; local buckling, strength, joints, fatigue, flexible modes, swept clearances and physical correlation remain unknown.

Actual revised mass propagates to axle reactions, inertia-aware 1.5/2/3 km propulsion/energy demand and crowd-sized concrete alternatives. Available motor performance and complete installed cost remain unknown. Improving a provisional target does not establish safe operation.

The planner diagnoses physical failures, advisory targets, numerical/model problems and missing evidence separately. Repeated unsuccessful local changes escalate toward new load paths or architecture. Pending work survives invocation budgets. Exact native physics can be reused under a new policy context without changing its original receipt or pretending a new solve occurred.

Inspect docs/design-engine/v06-adaptive-design/ and ./osr design adaptive plan/run/publish/verify. The Workbench exposes the variables and native comparisons; the current reference and city geography remain unchanged. The search continues through its explicit frontier rather than declaring that one failed comparison ends the design.

14. Reproduce and inspect

Use ./osr status, ./osr readiness, ./osr build, ./osr verify, ./osr test, ./osr book --check and ./osr workbench. ./osr sim baghdad reports its deliberate city-evaluation hold. ./osr legacy verify checks retained source/payload bytes at their recorded Git revisions.

The broader design-space comparison adds supplier-bound PET/PVC and directional honeycomb cores, aluminium/GFRP load paths, joints/inserts, motor topologies, glazing and HVAC leads. Run ./osr design space build/verify. Necessary bounds and component fixtures guide the next experiments; they do not establish a complete train or silently change the default.

The v0.5 archive preserves magnetic development evidence; its source and results are not overwritten to match a wheel-supported train. Historical numerical regeneration belongs in a separate checkout of the pinned revision.

Repository records

engineering/development-baseline.toml

engineering/station_rail/brief.toml

docs/design-engine/current/README.md

docs/design-engine/current/evaluation.json.gz

docs/design-engine/current/comparison.json

docs/design-engine/current/manifest.json

docs/design-engine/current/exports/manifest.json

docs/archives/v0.5-maglev/manifest.json

Physical and evaluation identities

selection id

8e5905bc7c81d6177bbaafde81ab3a5f 8cc54947c4923eb4493949566ac85cdc

candidate id

756a68870162f7559b052b05c9b20172 8c9f448e3bb61114f287848c02f2d3a3

body candidate id

9489e4f2666fd3b3c73232c86df3f12a df42a5531eb27373834cf3fecb1a3117

support component id

3e9aef846e6ec55bac01351216339461 8b37cf8c74b1100d6f8e5450ae5479d7

propulsion component id

67e2bfc4ef429a6c49e6d07095bc298c 4eb76b8386e4ee316c440ff128c30fa9

Source receipts bind the exact files used. Native method convergence, analytical screening, independent review and authenticated physical acceptance remain separate levels of evidence.